



---

# GEOPOLITICAL CONFLICTS & THE CRUDE OIL ECONOMY

Mapping Benchmark Volatility, Trade Rerouting,  
and Corporate Sourcing Strategies (2022–2026)

|                 |                   |
|-----------------|-------------------|
| Name :          | Joseph Biju       |
| Temporal Scope: | Q1 2022 – Q2 2026 |
| Enrollment :    | 23112047          |



**Primary Inquiry:** How do modern geopolitical conflicts alter the crude oil economy through supply expectations, transit bottlenecks, and sanctions—and how should exposed firms redesign their defense?



**Geopolitical Decoupling:**

Modern conflicts inject a persistent risk premium that separates oil prices from underlying physical supply/demand fundamentals.



**Rerouting Redirection:**

Avoidance of the Red Sea/Bab al-Mandab raises ton-mile demand, forcing an immediate restructure of global supply lines.



**Pragmatic Procurement:**

Emerging economies buffer domestic inflation by importing discounted, sanctioned crude under complex waiver frameworks.



**Strategic Hedging:**

Energy-intensive sectors must shift from reactive spot-buying to proactive, multi-layered risk management.



# Brent Crude Oil Price & Key Geopolitical Event Milestones (2022-2026)





## Physical Supply Shock

### Market Flow:

Actual loss of barrels (e.g., destroyed pipelines).

### Price Trajectory:

Sharp, structural, and long-term changes.

### Corporate Action:

Immediate physical rationing required.

## Sentiment Risk / Fear Premium

### Market Flow:

Oil continues flowing. Markets price the probability of future disruption via options trading.

### Price Trajectory:

Artificial, short-term spikes (e.g., April 2024 Iran-Israel strikes where zero oil was lost).

### Corporate Action:

Wait it out. Buying locks in inflated fear costs.



Geopolitical  
Event

```
graph LR; A[Geopolitical Event] --- B[1. Direct Production Threats: Destruction of physical wells or Gulf refineries.]; A --- C[2. Transit Chokepoints: Impairment of critical sea lanes (Strait of Hormuz, Bab al-Mandab).]; A --- D[3. Logistics & Shipping: Higher hull insurance, freight rate inflation, extended voyages.]; A --- E[4. Sanctions & Compliance: Financial constraints, G7 price caps, banking restrictions.]; A --- F[5. Risk Premium: Speculative options trading driving prices up before a single drop of oil is lost.]; B --- G[Inflated Corporate Cost]; C --- G; D --- G; E --- G; F --- G;
```

1. Direct Production Threats: Destruction of physical wells or Gulf refineries.

2. Transit Chokepoints: Impairment of critical sea lanes (Strait of Hormuz, Bab al-Mandab).

3. Logistics & Shipping: Higher hull insurance, freight rate inflation, extended voyages.

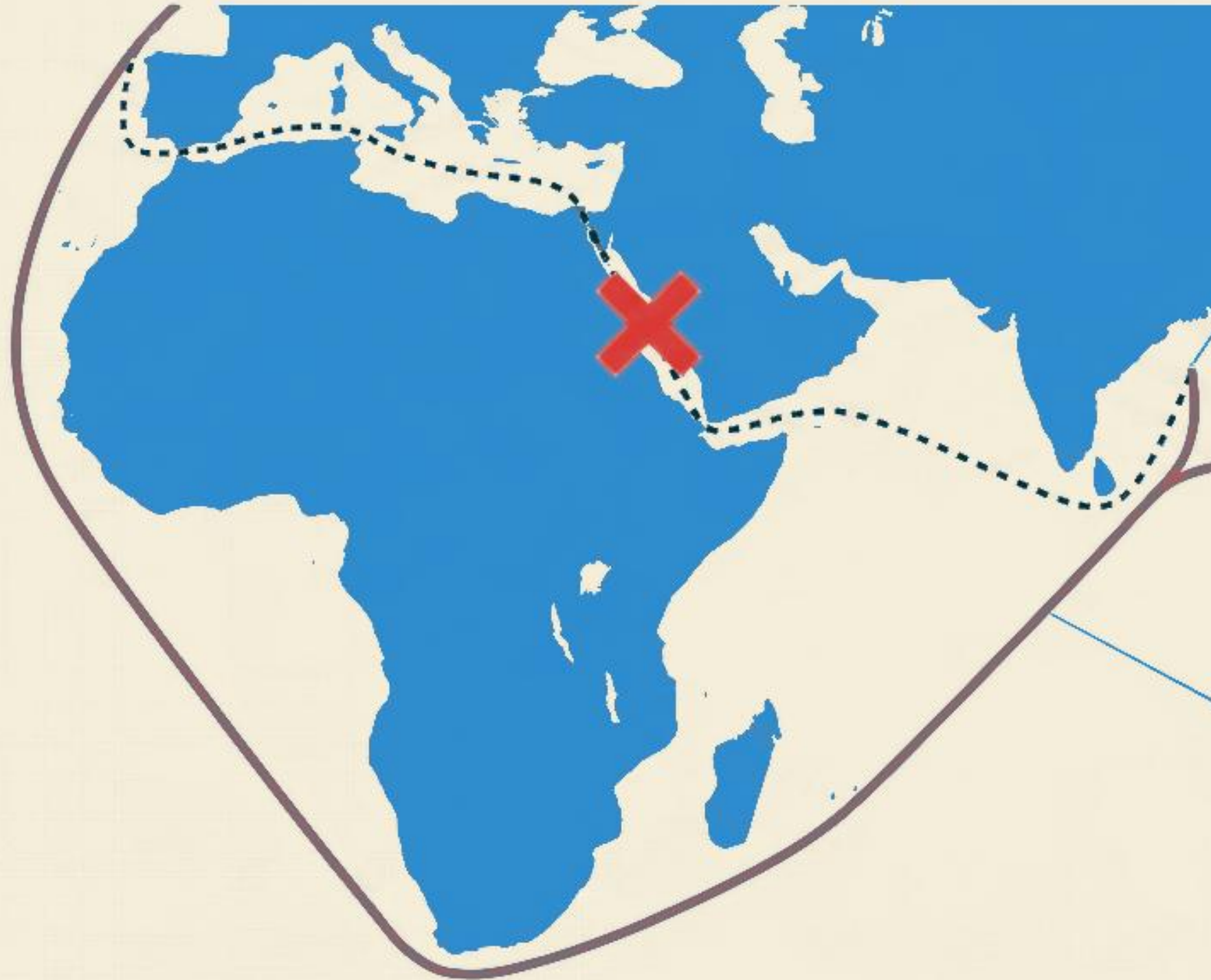
4. Sanctions & Compliance: Financial constraints, G7 price caps, banking restrictions.

5. Risk Premium: Speculative options trading driving prices up before a single drop of oil is lost.

Inflated  
Corporate Cost



# Anatomy of a Reroute



## The Ton-Mile Effect:

Longer routes soak up global tanker capacity, instantly inflating baseline freight rates globally.

## Time Penalty:

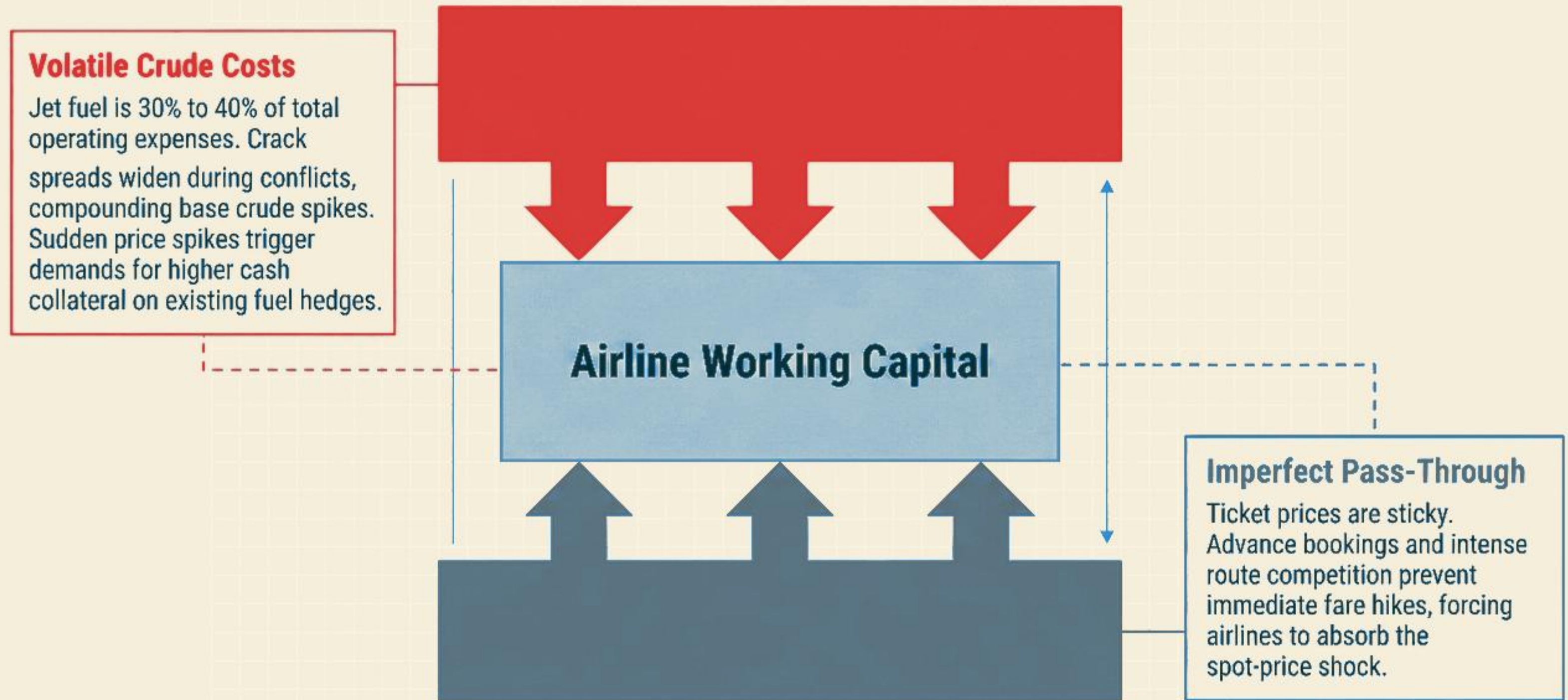
Adds 10 to 14 extra days to voyage times between Asia/Middle East and Europe.

## Cost Multipliers:

Drastic increase in fuel consumption per voyage + surging war-risk hull insurance premiums.

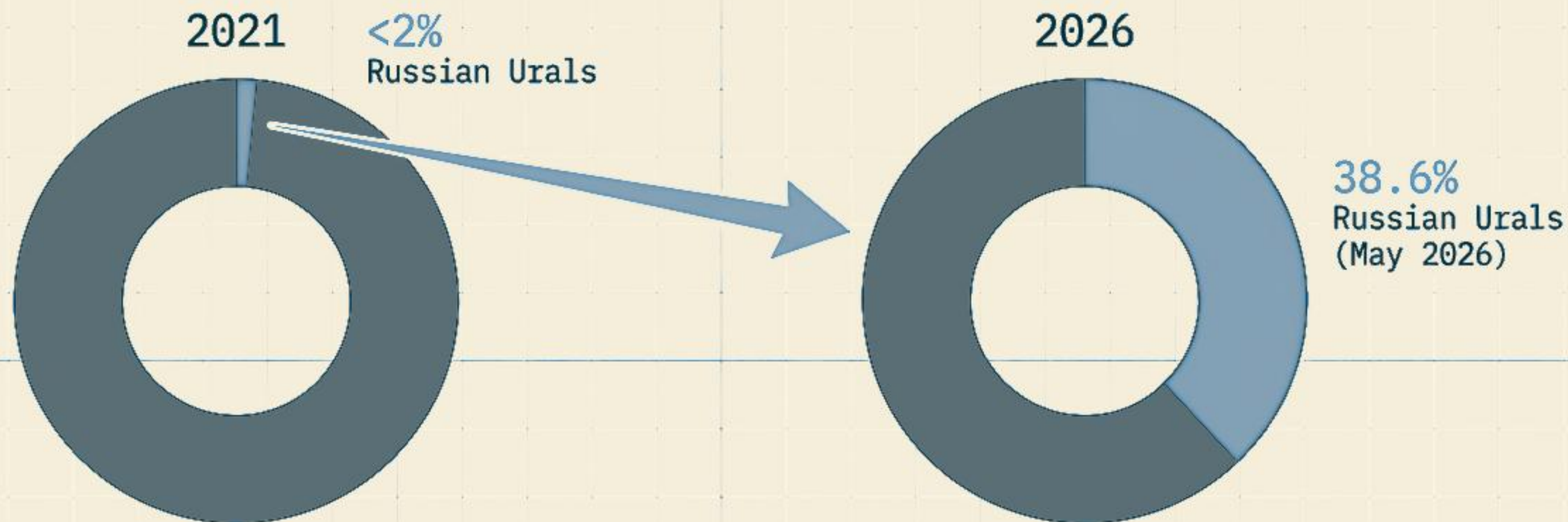


# Margin Vise: Architectural Cross-Section of Financial Pressure





# INDIA'S CRUDE OIL: SHARE OF WALLET SHIFT (2021 vs 2026)



**NATIONAL IMPERATIVE:** As the world's 3rd largest importer (85% dependent), energy security dictates Indian policy.

**THE SANCTION DISCOUNT:** Leveraging G7 waiver frameworks to import at discounts of \$20-\$30/bbl relative to Brent.

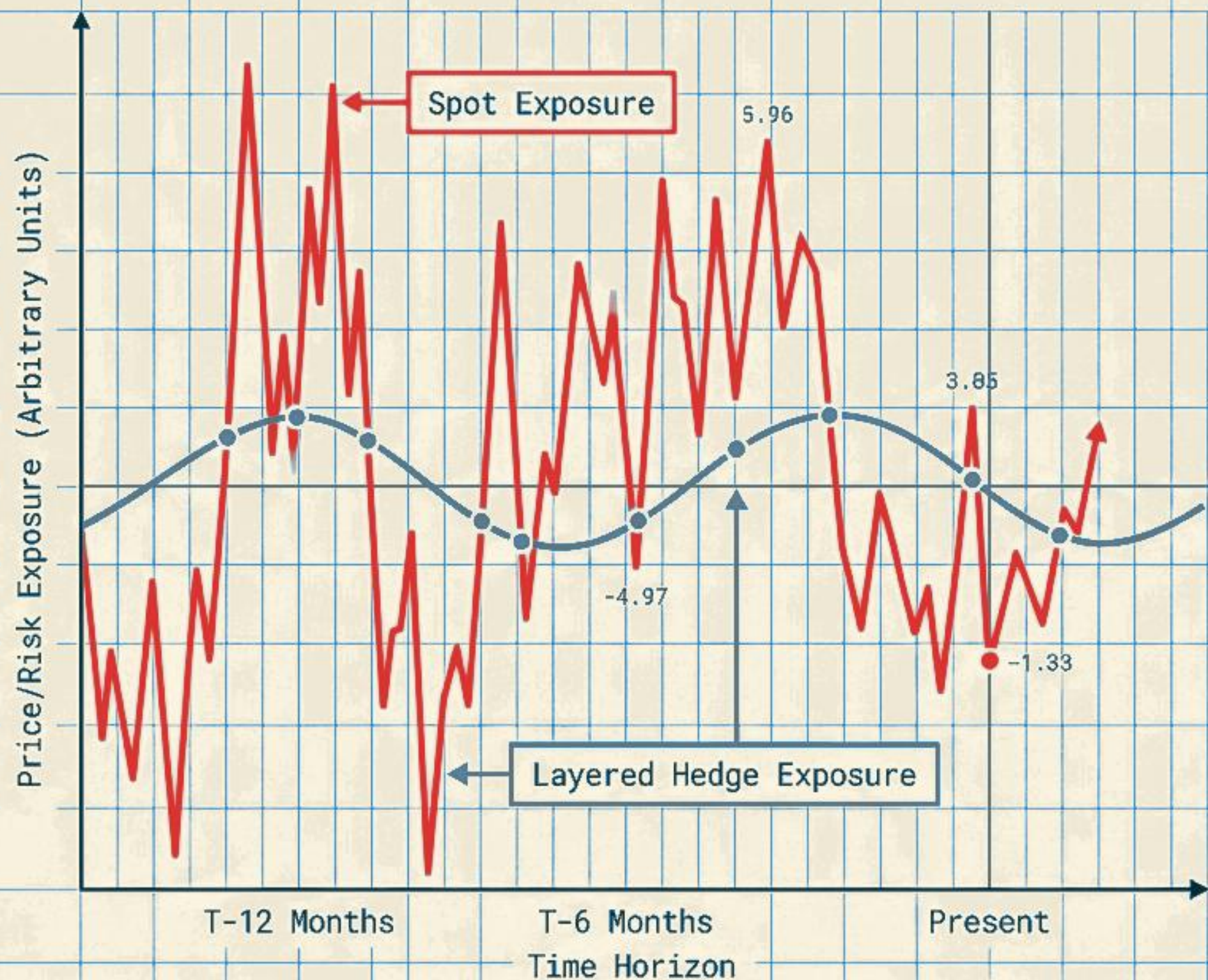
**THE PIVOT:** Russian Urals share surged from <2% (2021) to 33% (March 2026) and ultimately 38.6% (May 2026).

**THE OUTCOME:** Successfully insulated domestic refining margins and buffered the public from localized fuel inflation.



| INTERVENTION TOOL                     | SHORT-TERM BENEFIT                                             | LONG-TERM RISK                                                       | HISTORICAL PRECEDENT                |
|---------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------|
| Strategic Petroleum Reserve Releases. | Provides immediate, highly visible supply liquidity.           | Depletes national leverage for future, more severe crises.           | US SPR releases in 2022.            |
| Tax and Excise Cuts.                  | Instantly shields the end-consumer from inflation.             | Directly drains government fiscal revenue.                           | Indian domestic fuel tax cuts.      |
| Diplomatic Diversification.           | Ensures physical supply continuity during localized embargoes. | Requires long lead times to establish state-to-state term contracts. | India-Brazil, India-Gulf relations. |





## THE HEDGING PARADOX

Hedging is a risk-reduction mechanism, not a speculative profit center.

Locking in 100% of volume during a geopolitical fear spike is an expensive mistake.

## THE LAYERED SOLUTION

Systematic, rolling defense prevents catastrophic capital drains.

Best practice requires abandoning all-or-nothing bets in favor of layered tranches (e.g., executing 50% of volume at 12 months out, and 25% at 6 months out).





## THE REACTIVE BUYER

- Confuses risk-premiums with physical shortages.
- Panic-buys at the top of the market, locking in artificial fear costs.
- Relies on rigid, single-source feedstock contracts.
- Absorbs margin destruction due to delayed pricing pass-through.



## THE STRUCTURALLY RESILIENT BUYER

- Understands that resilience cannot be reactive; it must be structural.
- Utilizes rigorous, mathematically driven rolling hedging schedules.
- Invests heavily in dual-feedstock flexibility (switching Middle East/Urals grades).
- Embeds dynamic, automated fuel surcharges directly into downstream contracts.



# OPERATIONAL BLUEPRINT: THREE-STAGE RESILIENCE IMPLEMENTATION



## IMMEDIATE (DAYS 1-30)

Audit existing fuel hedging contracts to enforce strict rolling, layered schedules.

Embed automatic fuel volatility surcharges into all downstream supply agreements to fix the pass-through lag.

## MEDIUM-TERM (MONTHS 1-6)

Conduct rigorous audits of maritime logistics chains to identify hidden exposure to the Red Sea or Strait of Hormuz.

Increase physical onshore storage buffer capacity to absorb inevitable shipping delays.

## LONG-TERM (YEARS 1-3)

Fund capital expenditures for dual-feedstock refining capabilities to capitalize on shifting regional crude discounts.

Accelerate fleet transitions to energy-efficient assets and alternative fuels (e.g., Sustainable Aviation Fuel).



# Key Data Sources & References

## Brent Crude Spot Prices & Historical Volatility

- U.S. Energy Information Administration (EIA) Spot Prices ([eia.gov](https://eia.gov))
- St. Louis Fed (FRED) Economic Database - Series DCOILBRENTU ([fred.stlouisfed.org](https://fred.stlouisfed.org))

## Indian Crude Sourcing & Import Statistics

- Ministry of Petroleum & Natural Gas, India - PPAC ([ppac.gov.in](https://ppac.gov.in))
- Ministry of Commerce & Industry, India - Import Export Data Bank ([commerce.gov.in](https://commerce.gov.in))
- Shipment Tracking: Vortexa Analytics & Kpler Tanker Flow Databases

## Maritime Chokepoints & Logistics Economics

- EIA World Oil Transit Chokepoints Brief (Strait of Hormuz & Bab al-Mandab)
- UNCTAD Reports on Red Sea Navigation Disruptions (2024–2026)
- Freight Rate & Insurance Indices: Clarksons Research & Drewry

## Sanctions & Price Cap Compliance Frameworks

- G7 Price Cap Coalition Compliance Advisories (OFAC & EU DG FISMA)
- International Energy Agency (IEA) Monthly Oil Market Reports

## Aviation Industry Transmission & Hedging Strategy

- IATA Jet Fuel Price Monitor & Refined Kerosene Crack Spreads ([iata.org](https://iata.org))
- Corporate Disclosures: InterGlobe Aviation (IndiGo) & Delta Air Lines Annual 10-K Filings

## Playbook Code & Data Repository

- GitHub Repository: [github.com/jozee-io/oil-geopolitics-2022-2026](https://github.com/jozee-io/oil-geopolitics-2022-2026)